

p. 567 [555] (tail of paper on a system of cubic equations)

$a^2d - 3abc + 2b^3$; the general value of u is $u =$ any function whatever of the last-mentioned three functions. We have to find the rational non-integral functions of these functions which are rational and integral functions of the coefficients; such a function is

$$\frac{1}{a^2} \{ (a^2d - 3abc + 2b^3) + \frac{1}{4}(ac - b^2)^3 \} = ac^3 + 4ac^2 + 4b^3d - 3b^2c^2 - 6abcd,$$

and the original three solutions, together with the last-mentioned function $a^2d^2 + \&c.$, constitute the complete system of the seminvariants of the cubic function; viz. every other seminvariant is a rational *and integral* function of these. And so, in the general case, the problem is to complete the series of the n solutions $a, ac - b^2, a^2d - 3abc + 2b^3, a^3e - 4a^2bd + 6ab^2c - 3b^4$, &c. by adding thereto the solutions which, being rational but non-integral functions of these, are rational and integral functions of the coefficients; and thus to arrive at a series of solutions such that every other solution is a rational and integral function of these.

4. *Note on certain Families of Surfaces.* Report, 1871, pp. 19, 20.

See the paper numbered 526, of which this Note is a duplicate.

5. *On the Mercator's Projection of a Surface of Revolution.* Report, 1873, p. 9.

THE theory of Mercator's projection is obviously applicable to any surface of revolution; the meridians and parallels are represented by two systems of parallel lines at right angles to each other, in such wise that for the infinitesimal rectangles included between two consecutive arcs of meridian and arcs of parallel the rectangle in the projection is similar to that on the surface. Or, what is the same thing, drawing on the surface the meridians at equal infinitesimal intervals of angular distance, we may draw the parallels at such intervals as to divide the surface into infinitesimal squares; the meridians and parallels are then the projections represented by two systems of equidistant parallel lines dividing the plane into squares. And if the angular distance between two consecutive meridians instead of being infinitesimal is taken moderately small (5° or even 10°), then it is easy on the surface or *in plano*, using only the curve which is the meridian of the surface, to lay down graphically the series of parallels which are in the projection represented by equidistant parallel lines. The method is, of course, an approximate one, by reason that the angular distance between the two consecutive meridians is finite instead of infinitesimal.

I have in this way constructed the projection of a skew hyperboloid of revolution; viz. in one figure I show the hyperbola, which is the meridian section, and by means of it (taking the interval of the meridians to be $= 10^\circ$) construct the positions of the

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successive parallels; I complete the figure by drawing the hyperbolas which are the orthographic projections of the meridians, and the right lines which are the orthographic projections of the parallels; the figure thus exhibits the orthographic projection (on the plane of a meridian) of the hyperboloid divided into squares as above. The other figure, which is the Mercator's projection, is simply two systems of equidistant parallel lines dividing the paper into squares. I remark that in the first figure the projections of the right lines on the surface are the tangents to the bounding hyperbola; in particular, the projection of one of these lines is an asymptote of the hyperbola. This I exhibit in the figure, and by means of it trace the Mercator's projection of the right line on the surface; viz. this is a serpentine curve included between the right lines which represent two opposite meridians and having these lines for asymptotes. It is sufficient to draw one of these curves, since obviously for any other line of the surface belonging to the same system the Mercator's projection is at once obtained by merely displacing the curve parallel to itself; and for any line of the other system the projection is a like curve in a reversed position.

A Mercator's projection might be made of a skew hyperboloid not of revolution; viz. the curves of curvature ought be drawn so as to divide the surface into squares, and the curves of curvature be then represented by equidistant parallel lines as above; and the construction would in only a little more difficult. The projection presented itself to me as a convenient one for the representation of the geodesic lines on the surface, and for exhibiting them in relation to the right lines of the surface; but I have not yet worked this out. In conclusion, it may be remarked that a surface in general cannot be divided into squares by its curves of curvature, but that it may be in an infinity of ways divided into squares by two systems of curves on the surface, and any such system of curves gives rise to a Mercator's projection of the surface.

NOTES AND REFERENCES.

518. BERTRAND, *C. R.*, t. LXXV. (1872), pp. 535–536, referring to my Note remarks that the condition can be (by means of the imaginary coordinates of M. Ossian Bonnet) expressed in a simple form communicated by him to the Philosophic Society, May 1870. I reproduce this investigation, although it is not easy to present it in a quite intelligible form. We take $p = f(x, y, z)$ to represent a family of surfaces belonging to a triply orthotomic system, and consider two neighbouring surfaces (A) and (A') corresponding to the values x and $x + dx$; A and A' the two points where they meet the trajectories of the surfaces; AT , $A'T'$ the tangents to the curves of curvature of the same system at A , A' respectively. Then according to the remark of M. Lévy, it is to be expressed that these lines meet, and this is done by expressing that along the trajectory AA' , the variation of the angle of AT with the osculating plane measures is equal to the angle of the osculating plane at A , A' respectively.

Let R be the spherical image of AT with RR , R the angle of AT with RR , $\theta - \theta$ is the angle of AT with the osculating plane at A of the trajectory: $d\theta - d\theta = d\theta_1$. Introducing the symmetric imaginary coordinates α and β , we have

$$\alpha = \frac{q}{\sqrt{q^2 - p^2}}, \quad \beta = \frac{r}{\sqrt{q^2 - p^2}}, \quad 1 = \frac{p}{\sqrt{q^2 - p^2}}; \quad d\beta = -\alpha d\beta, \quad d\alpha = \beta d\alpha.$$

But $d\alpha$ and $d\beta$ being the increments of α , β corresponding to $d\alpha$ in the passage from A to A' , then by a theorem of M. Liouville

$$d\eta = \alpha d\beta \left(\frac{d\beta}{d\alpha} da - \frac{d\beta}{d\alpha} d\beta \right);$$

the condition then is

$$d\theta = \alpha \left(\frac{d\beta}{d\alpha} da - \frac{d\beta}{d\alpha} d\beta \right).$$

and the formula

$$e^{ix d\xi} = b \sqrt{\frac{da}{d\beta}} \cdot \sqrt{\frac{d\beta}{d\eta}},$$

enables this to be written in the definitive form

$$d\alpha \frac{d}{d\eta} \left\{ b \sqrt{\frac{db}{d\beta}} \cdot \frac{d\eta}{d\xi} - \frac{d\eta}{d\beta} \right\} + d\beta \frac{d}{d\xi} \left\{ b \sqrt{\frac{db}{d\beta}} \cdot \frac{d\eta}{d\beta} \right\} + d\delta \left[\frac{d}{d\delta} \left(\frac{d\eta}{d\xi} \right) - \frac{d}{d\delta} \left(\frac{d\eta}{d\beta} \right) \right] = 0.$$

We have

$$\begin{aligned} dx \left(\frac{1}{2}p + s \right) + dy \frac{db}{d\eta} d\beta + b \frac{db}{d\xi} &= 0, \\ dx \frac{d\eta}{d\xi} + dy \left(\frac{1}{2}p + s \right) + dz \frac{d\eta}{d\beta} &= 0, \end{aligned}$$

and thence eliminating dx, dy, dz , we have

$$\begin{vmatrix} \frac{d}{d\eta} \left(b \sqrt{\frac{db}{d\beta}} \cdot \frac{d\eta}{d\xi} - \frac{d\eta}{d\beta} \right) & \frac{d}{d\xi} \left(b \sqrt{\frac{db}{d\beta}} \cdot \frac{d\eta}{d\beta} \right) & \frac{d}{d\delta} \left(\frac{d\eta}{d\xi} \right) - \frac{d}{d\delta} \left(\frac{d\eta}{d\beta} \right) \\ \frac{1}{2}p + s & \frac{db}{d\eta} & \frac{db}{d\xi} \\ \frac{d\eta}{d\xi} & \frac{1}{2}p + s & \frac{d\eta}{d\beta} \end{vmatrix} = 0,$$

which defines the triply orthotomic system.

END OF VOL. VIII.